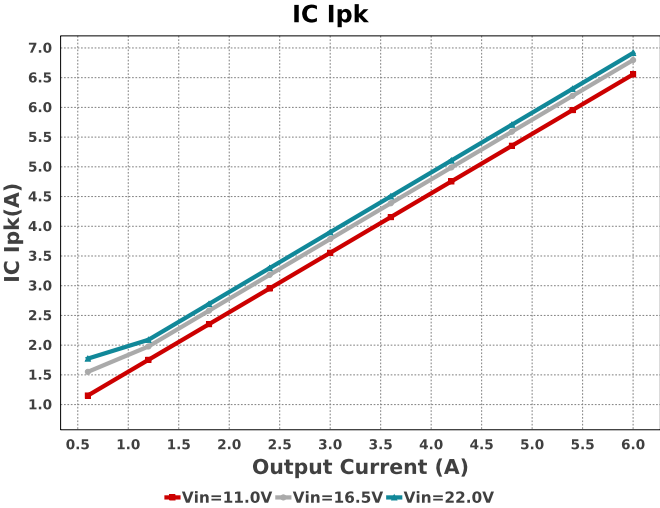
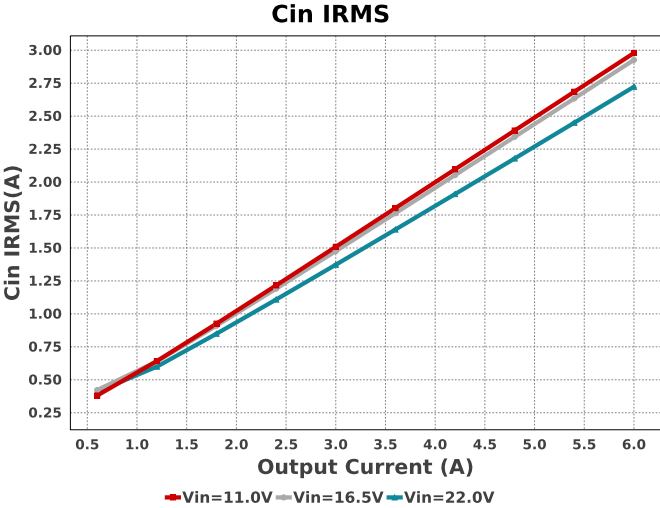
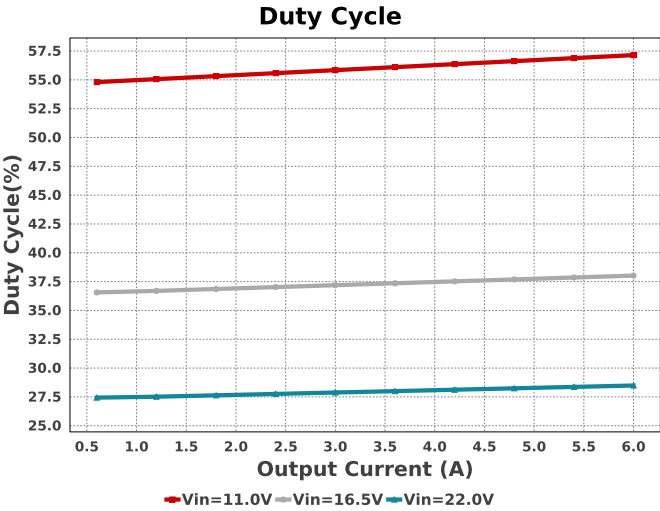
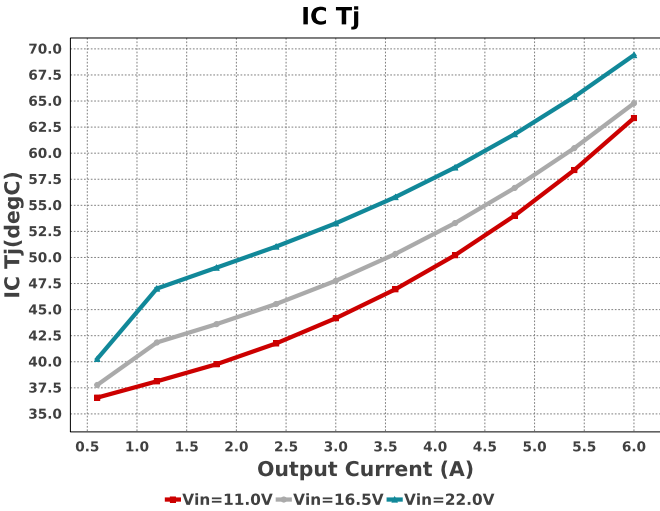
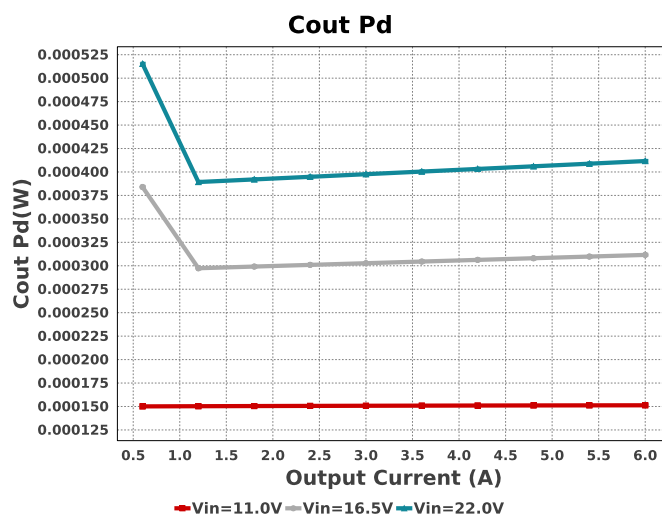
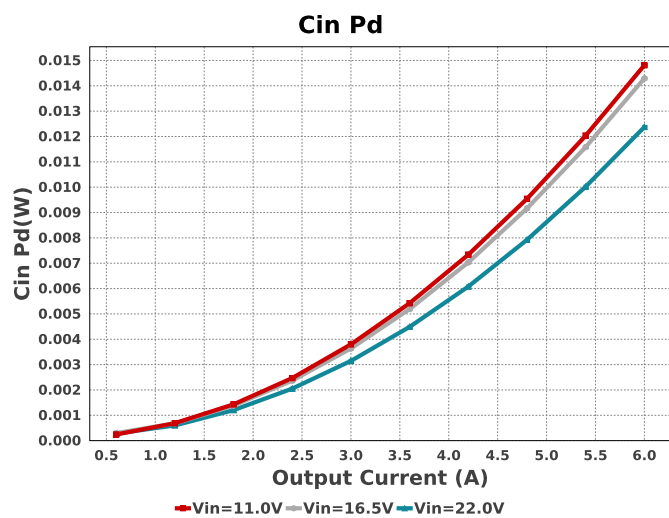
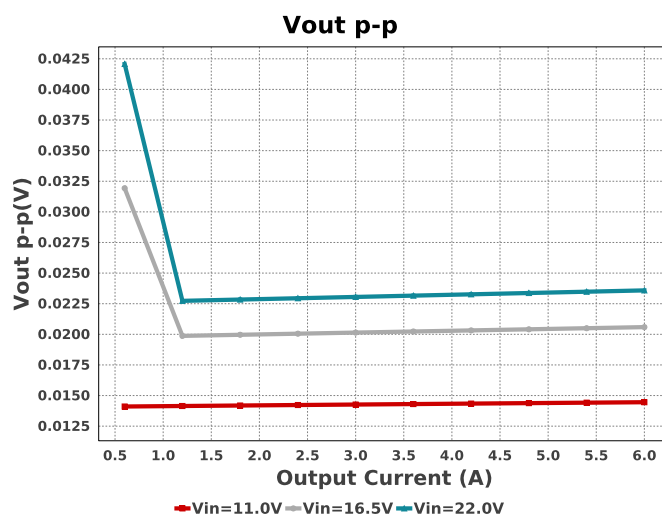
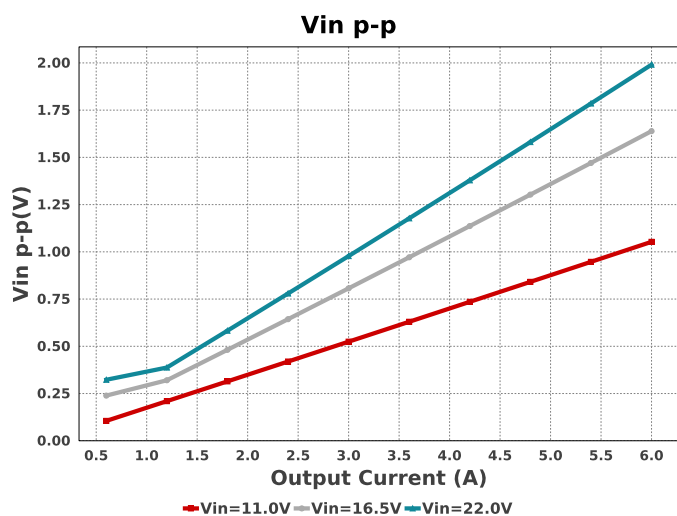
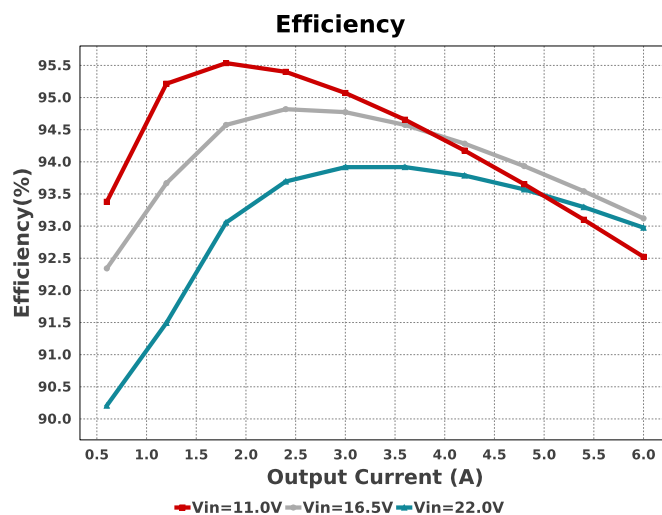
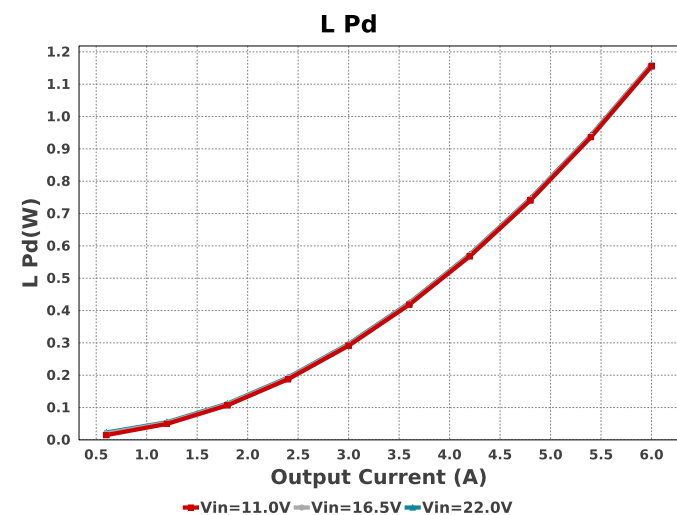




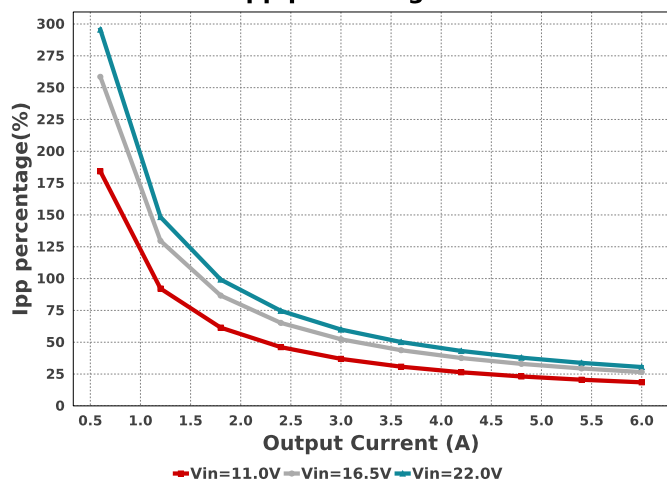
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Topology = Buck
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BOM Cost = \$3.85
BOM Count = 19
Total Pd = 2.71W

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rfb	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfbt	Panasonic	ERJ-6ENF9092V Series= ERJ-6E	Res= 90.9 kOhm Power= 125.0 mW Tolerance= 1.0%	1	\$0.01	0805 7 mm ²
Rmode	Vishay-Dale	CRCW040252K3FKED Series= CRCW..e3	Res= 52.3 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rpg	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
U1	Texas Instruments	TPS56A37RPAR	Switcher	1	\$1.11	RPA0010A 16 mm ²

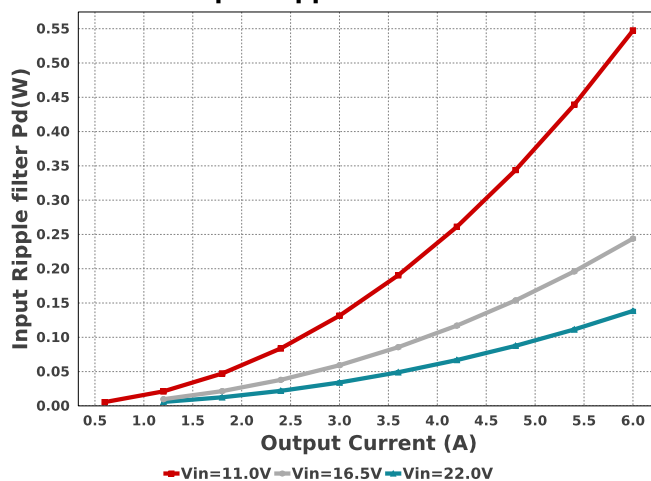




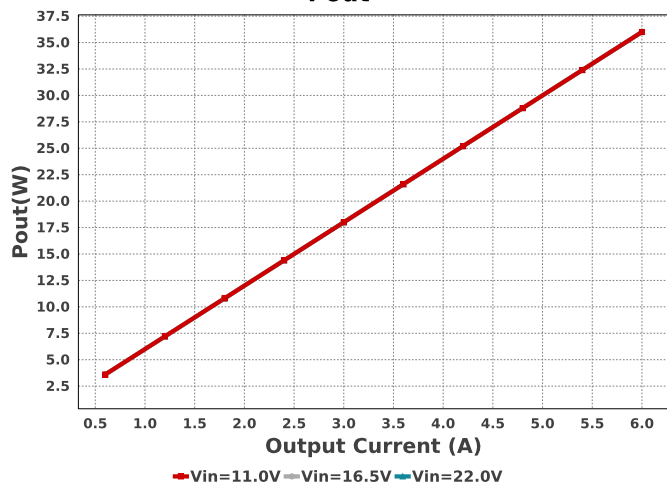
Ipp percentage



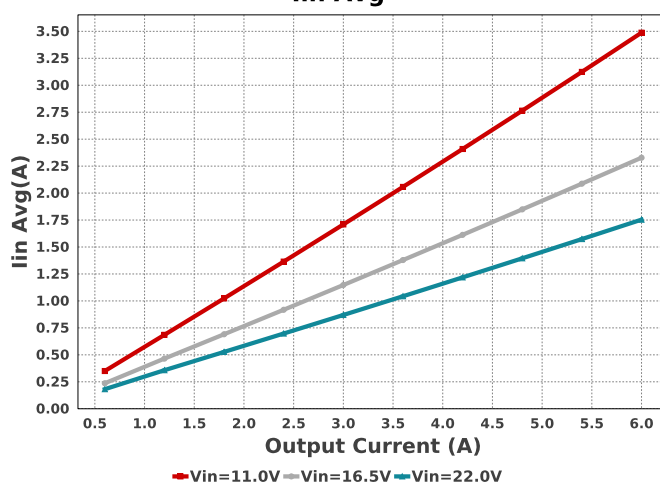
Input Ripple filter Pd



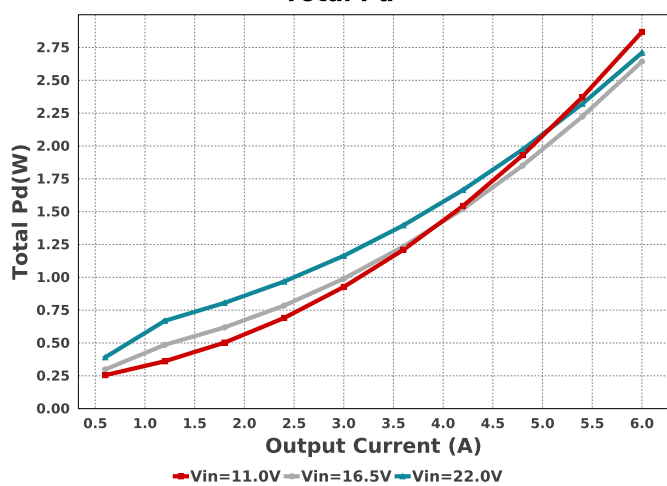
Pout



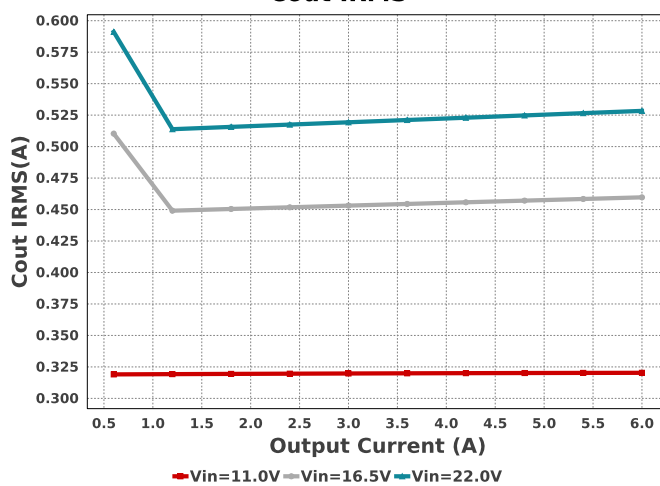
Iin Avg

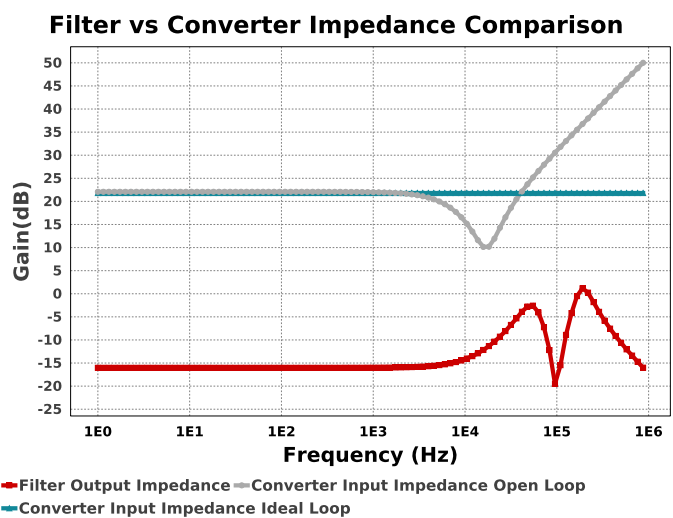
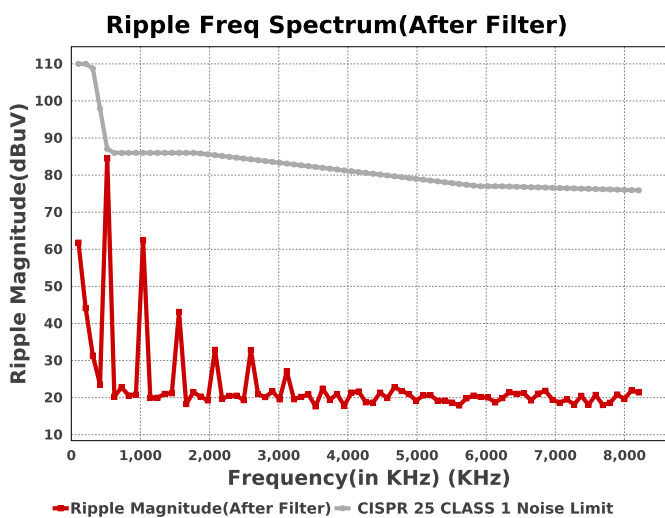
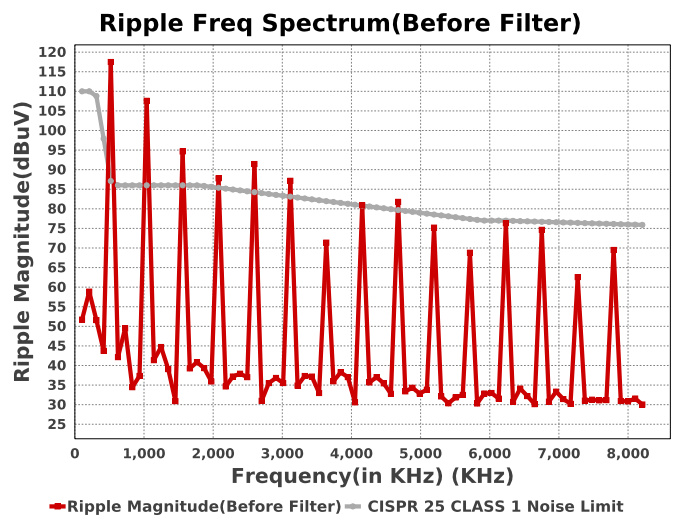
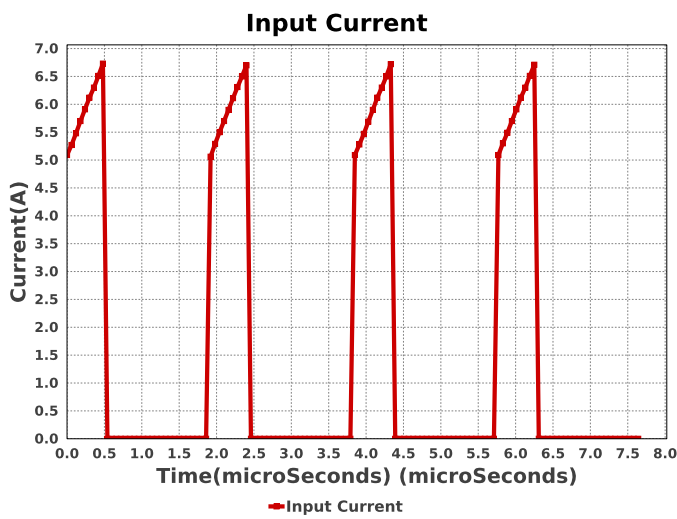
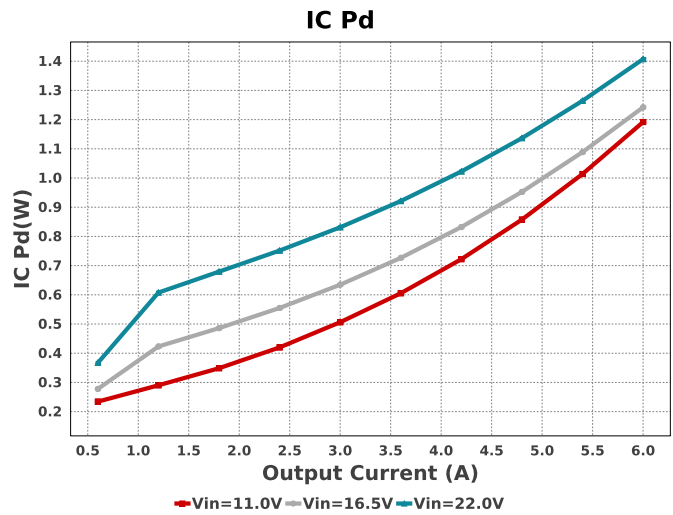
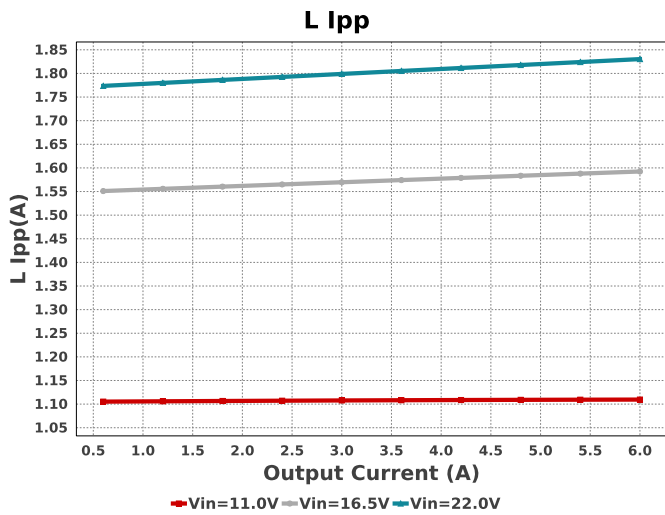


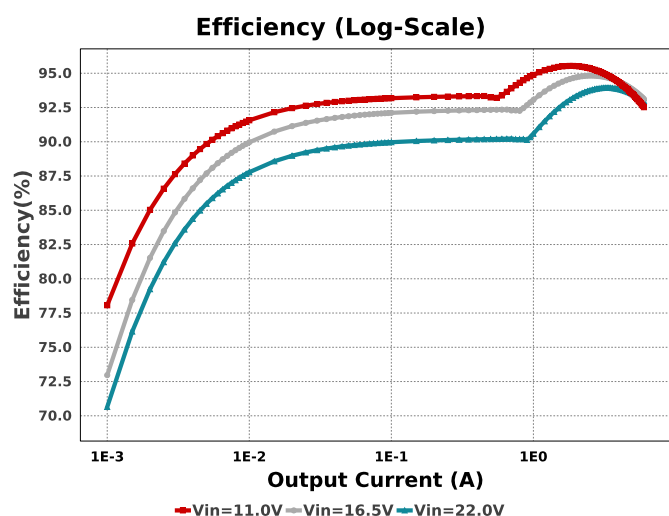
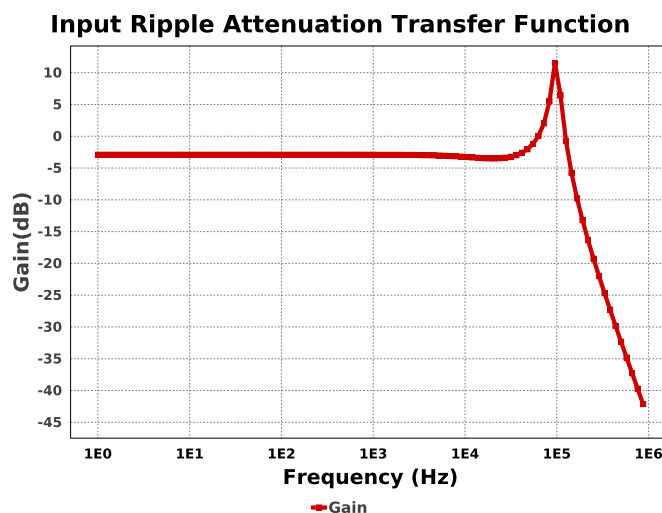
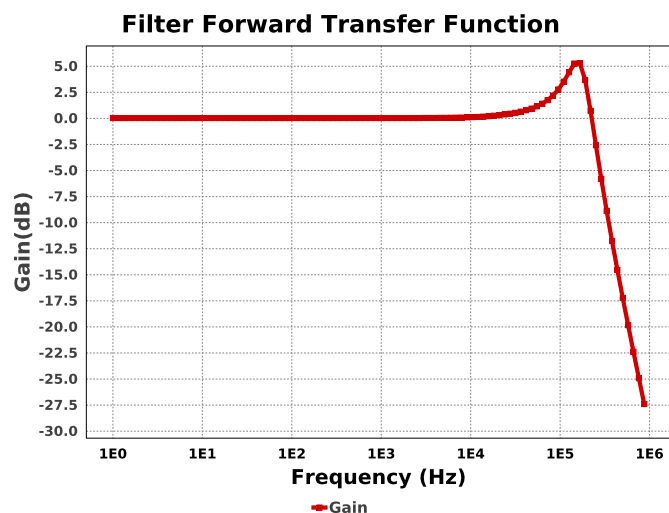
Total Pd



Cout IRMS







Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	2.723 A	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	12.373 mW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	528.371 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	411.65 μ W	Capacitor	Output capacitor power dissipation
5.	Input Ripple Noise After input filter	84.49 dBuV	EMI Noise	Input Ripple Noise after filter at switching frequency
6.	Input Ripple Noise before input filter	117.5 dBuV	EMI Noise	Input Ripple Noise before filter at switching frequency
7.	Input Ripple filter Pd	138.4 mW	EMI Noise	Input Ripple Filter Power Dissipation
8.	Noise limits defined by CISPR Standards	87.1 dBuV	EMI Noise	Noise limits for CLASS 1 of CISPR 25 standard
9.	IC Ipk	6.915 A	IC	Peak switch current in IC
10.	IC Pd	1.407 W	IC	IC power dissipation
11.	IC Tj	69.403 degC	IC	IC junction temperature
12.	IC Tolerance	6.0 mV	IC	IC Feedback Tolerance
13.	ICThetaJA Effective	28.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
14.	Iin Avg	1.754 A	IC	Average input current
15.	Ipp percentage	30.506 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
16.	L Ipp	1.83 A	Inductor	Peak-to-peak inductor ripple current
17.	L Pd	1.161 W	Inductor	Inductor power dissipation
18.	Cin Pd	12.373 mW	Power	Input capacitor power dissipation
19.	Cout Pd	411.65 μ W	Power	Output capacitor power dissipation
20.	IC Pd	1.407 W	Power	IC power dissipation
21.	Input Ripple filter Pd	138.4 mW	Power	Input Ripple Filter Power Dissipation
22.	L Pd	1.161 W	Power	Inductor power dissipation
23.	Total Pd	2.71 W	Power	Total Power Dissipation
24.	BOM Count	19	System Information	Total Design BOM count
25.	Duty Cycle	28.487 %	System Information	Duty cycle

#	Name	Value	Category	Description
26.	Efficiency	92.976 %	System Information	Steady state efficiency
27.	FootPrint	259.0 mm ²	System Information	Total Foot Print Area of BOM components
28.	Frequency	519.418 kHz	System Information	Switching frequency
29.	Iout	6.0 A	System Information	Iout operating point
30.	Mode	CCM	System Information	Conduction Mode
31.	Pout	36.0 W	System Information	Total output power
32.	Total BOM	\$3.85	System Information	Total BOM Cost
33.	Vin	22.0 V	System Information	Vin operating point
34.	Vin p-p	1.991 V	System Information	Peak-to-peak input voltage
35.	Vout	6.0 V	System Information	Operational Output Voltage
36.	Vout Actual	6.054 V	System Information	Vout Actual calculated based on selected voltage divider resistors
37.	Vout Tolerance	2.838 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
38.	Vout p-p	23.587 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	6.0	Maximum Output Current
VinMax	22.0	Maximum input voltage
VinMin	11.0	Minimum input voltage
VinTyp	14.8	Typical input voltage
Vout	6.0	Output Voltage
base_pn	TPS56A37	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

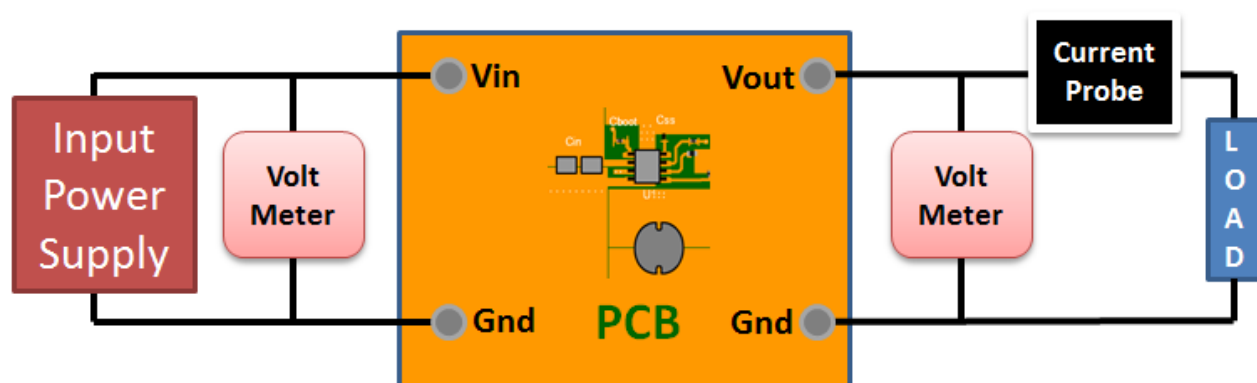
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 11.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 4FF96F465380DFD3DEA23FCB45ABC5B8[v1]
2. **TPS56A37** Product Folder : <http://www.ti.com/product/TPS56A37> : contains the data sheet and other resources.

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